REFRIGERANT SYSTEM

id071100000900

Outline

- The refrigerant system cools the evaporator using the cooling cycle to cool and dehumidify the air passing through the evaporator.

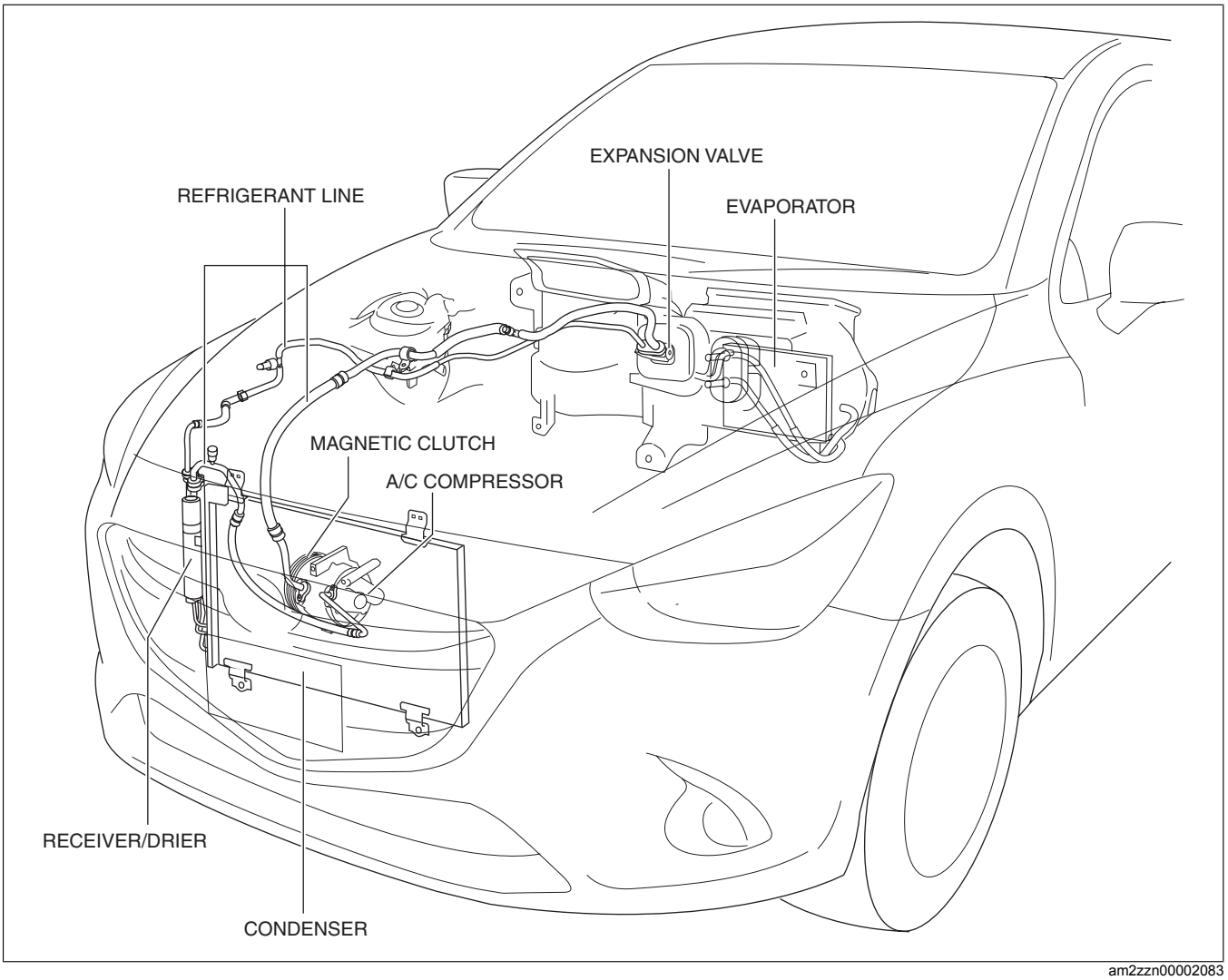
Function

- The refrigerant system changes the pressure and temperature of the charged refrigerant and cools the evaporator at the refrigerant cycle in which the refrigerant is liquefied or atomized.

Structural View

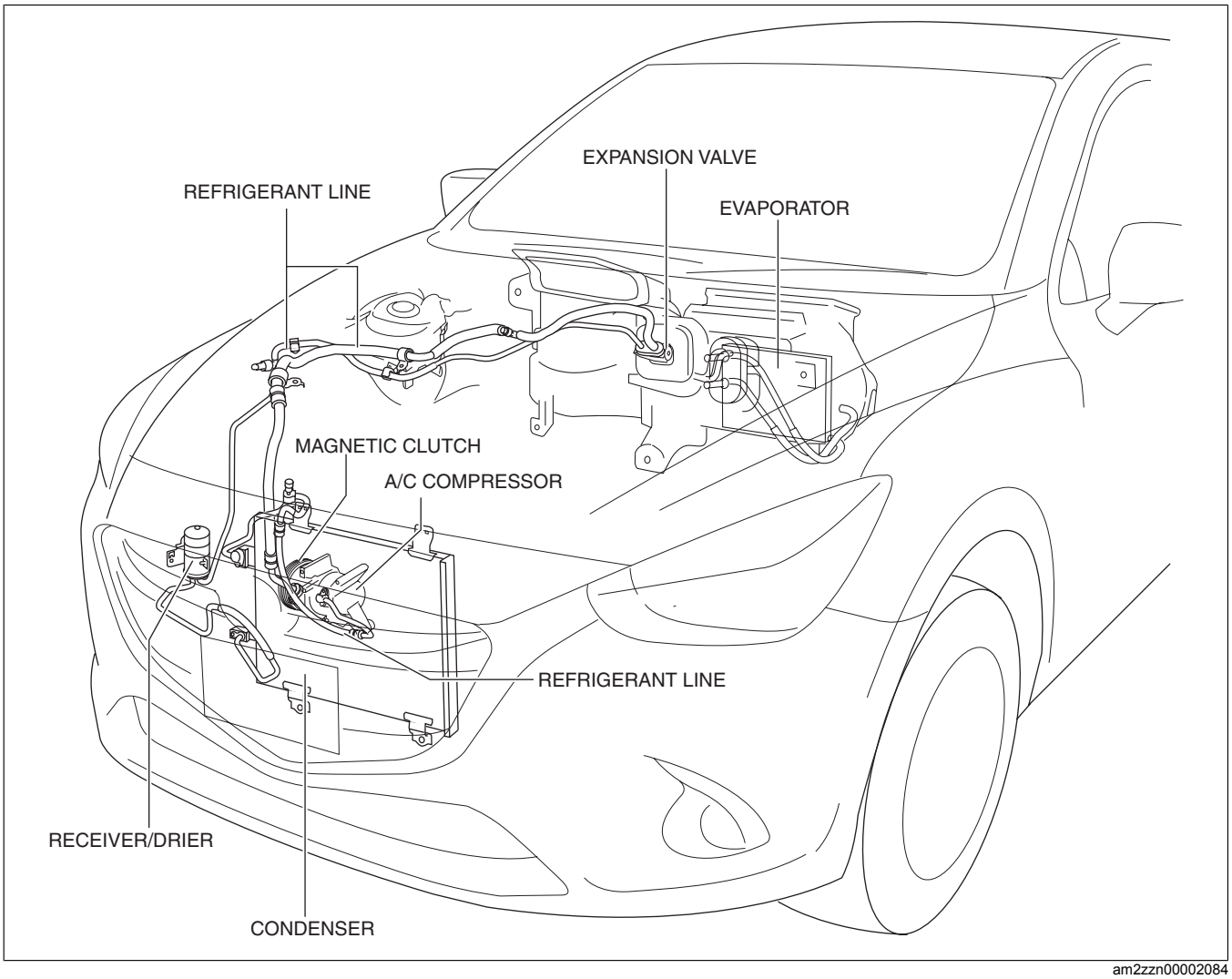
- The refrigerant system consists of the following parts.

SKYACTIV-G 1.3, SKYACTIV-G 1.5



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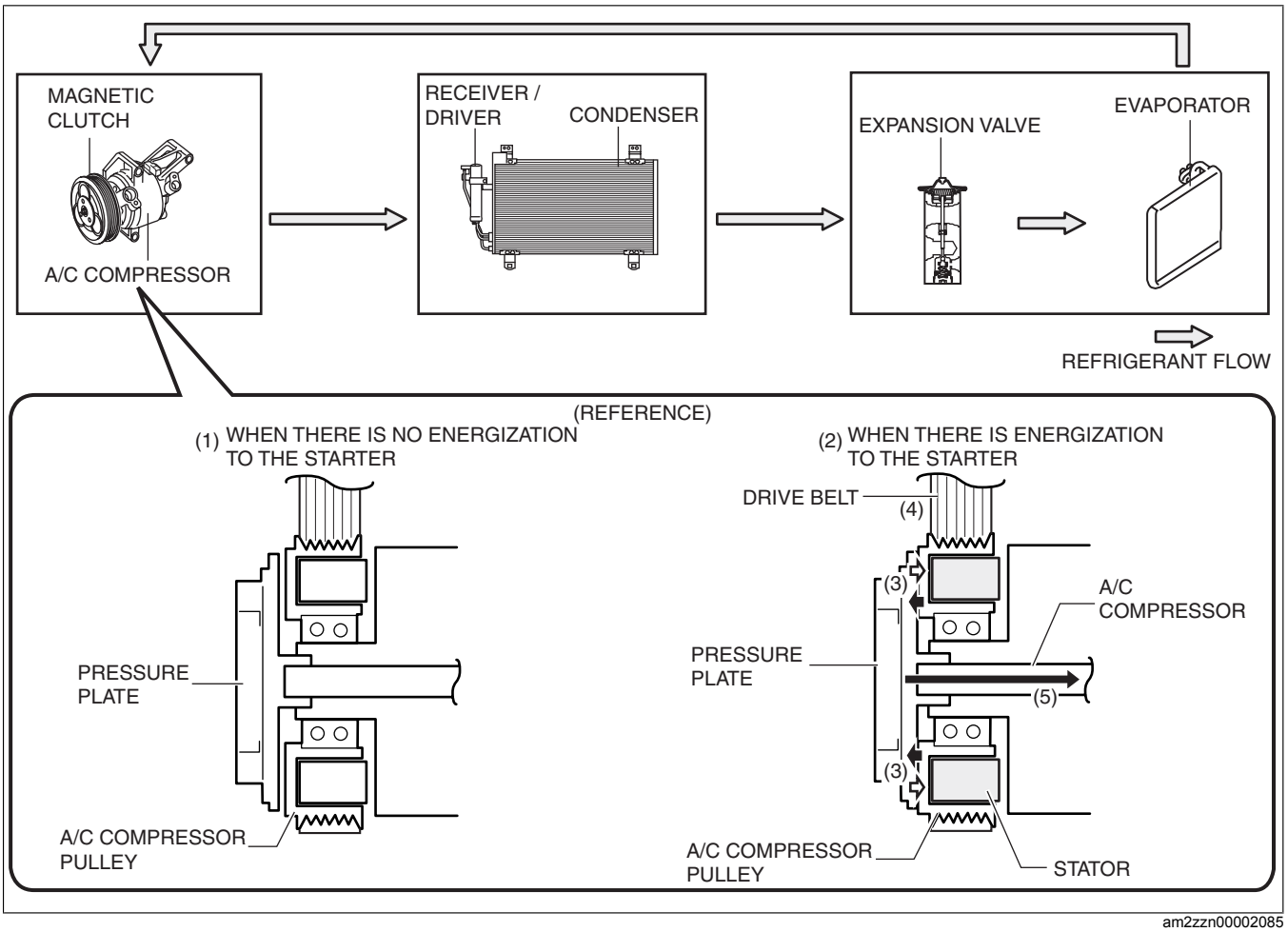
SKYACTIV-D 1.5



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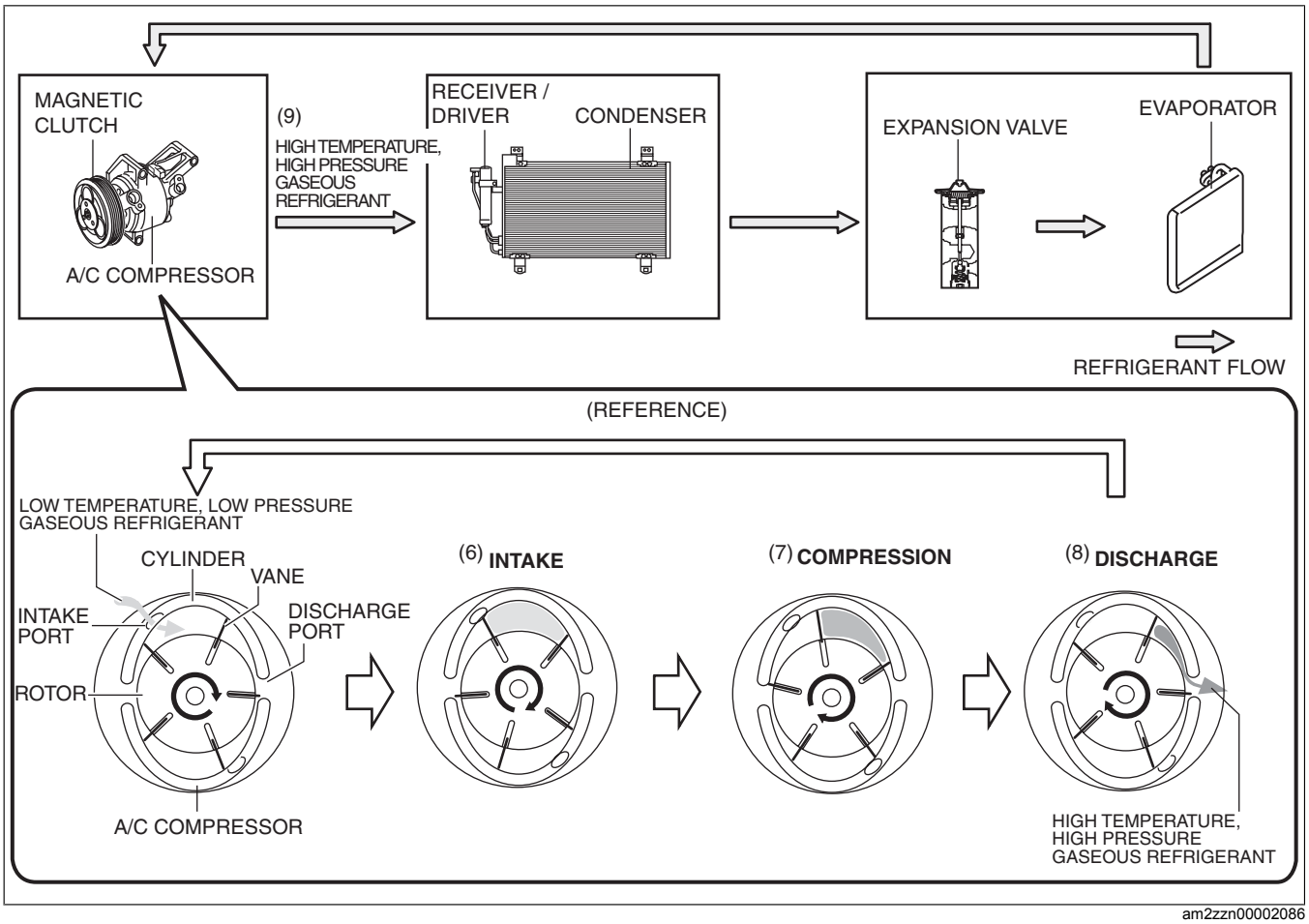
Operation

1. When there is no energization to the starter (A/C relay off) (1), the pressure plate and A/C compressor pulley do not engage.
2. When there is energization to the starter (A/C relay on) (2), the magnetic clutch of the starter operates and the pressure plate and A/C compressor pulley engage (3).
3. The engine power (4) from the drive belt is transmitted (5) to the A/C compressor via the pressure plate from the A/C compressor pulley.



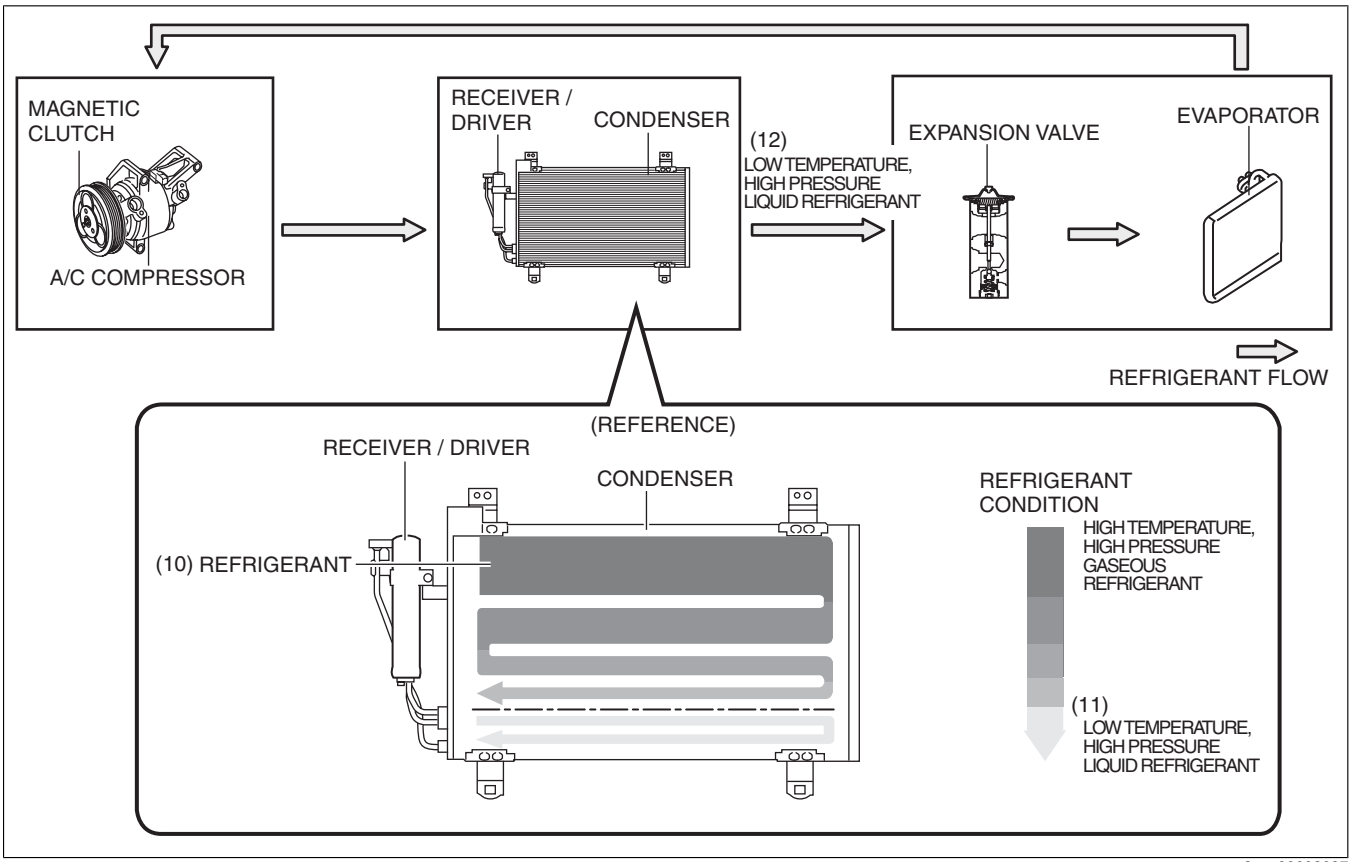
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4. When the magnetic clutch engages, the A/C compressor is driven.
5. The A/C compressor intakes (6) low temperature, low pressure gaseous refrigerant.
6. The A/C compressor pressurizes (7) the gaseous refrigerant in the cylinder by the rotation of the rotor.
7. Because the condensing temperature is low, the low pressure gaseous refrigerant is pressurized by the A/C compressor, which raises the condensing temperature so that the gaseous refrigerant condenses when it is cooled at the condenser.
8. The A/C compressor discharges (8) the high temperature, high pressure gaseous refrigerant by the rotation of the rotor.
9. The high temperature, high pressure gaseous refrigerant is sent (9) to the condenser.



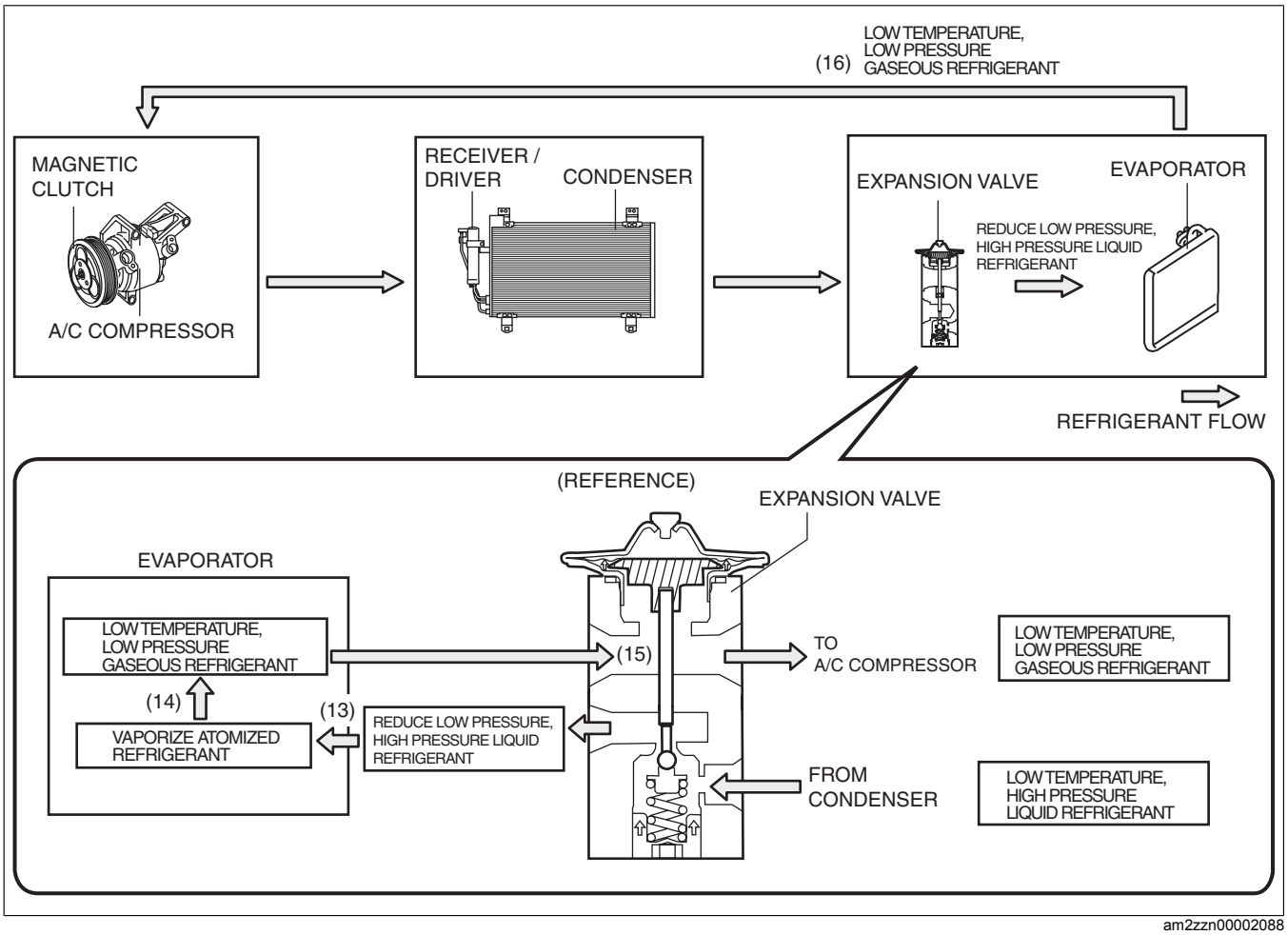
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10. The high temperature, high pressure gaseous refrigerant is circulated (10) inside the side of the condenser.
11. When the airflow from the exterior contacts the condenser fins, the heat in the circulating refrigerant is released through heat exchange, and the once high temperature, high pressure gaseous refrigerant becomes a low temperature, high pressure liquid refrigerant.
12. The low temperature, high pressure liquid refrigerant is purified and dehumidified by the receiver/drier and sent to the expansion valve (12).



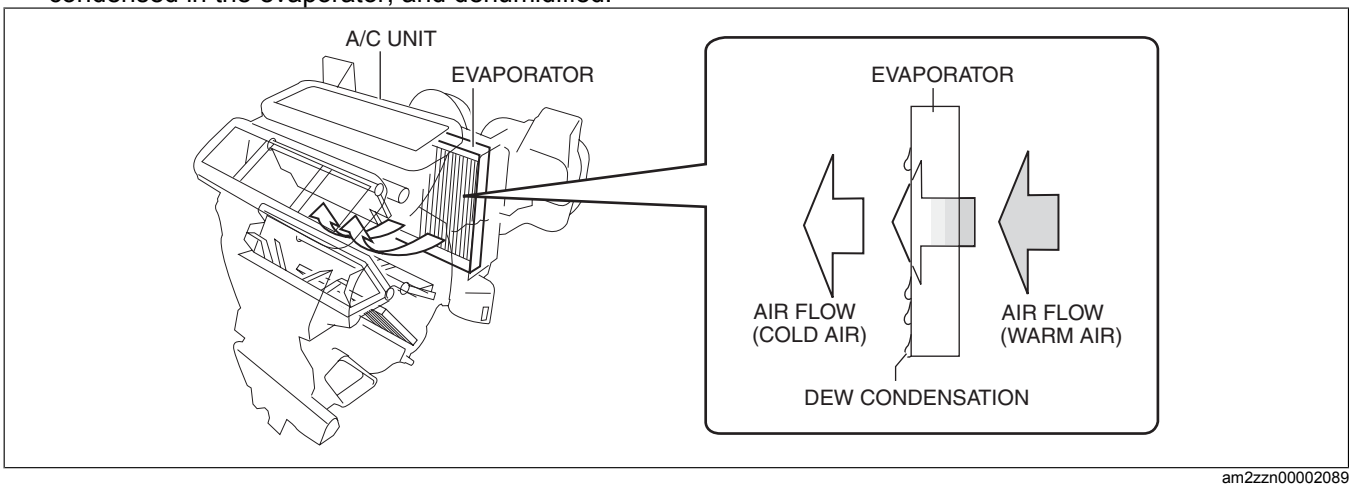
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13. The low temperature, high pressure liquid refrigerant is depressurized by the expansion valve and converted to a mist which is sent (13) to the evaporator.
14. The liquid refrigerant, now in a mist state, is atomized in the evaporator and becomes (14) a low temperature, low pressure gaseous refrigerant, and the evaporator is cooled by the surrounding heat being drawn out (latent heat) when the liquid is atomized.
15. The expansion valve adjusts the liquid refrigerant flow volume to the evaporator according to the gaseous refrigerant temperature (15) circulating from the evaporator to the condenser.
16. The low temperature, low pressure gaseous refrigerant is sent to the A/C compressor and circulates (16) the refrigerant cycle.



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17. The airflow passing through the cooled evaporator is cooled via heat exchange, the moisture in the air is condensed in the evaporator, and dehumidified.



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